

Dissertation

# **Non-Parametric Reconstruction of the Hubble Parameter with LISA using Gaussian Process Regression**

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# *Abstract*

## **Non-Parametric Reconstruction of the Hubble Parameter with LISA using Gaussian Process Regression**

by Krishnanjan SIL

The present dissertation is devoted to the model-independent, non-parametric reconstruction of the Hubble parameter,  $H(z)$ , using simulated “standard siren” data from the future gravitational-wave observatory: LISA, adopting the Gaussian Process regression (GPR). Here, I have assumed the Vanilla  $\Lambda$ CDM model of the universe. I have also generated the  $d_L$  vs  $z$  plot for another future GW mission: the Einstein Telescope (ET), the GP reconstruction part for it has been kept for future research due to high computation time and suspected errors from computing derivatives numerically rather using properties of GP kernels to do so. After reviewing direct (HST/Cepheids) and indirect (SNe Ia, CMB, BAO) measurements of  $H_0$ , I believe standard sirens as a calibration-free probe. I have generated realistic mock catalogs for LISA and ET, applied Gaussian Process regression (GPR) to reconstruct  $H(z)$  for LISA, and cross-checked against noise-free data. For LISA, the GPR yields stable results up to  $z = 4$ . Here, I have also discussed about alternatives to GPR such as Artificial Neural Networks (ANN) to reconstruct  $d_L$  for LISA in a larger range, up to  $z = 9$ . Key challenges include GaPP compatibility with Python 3, derivative errors from *np.gradient*, and limited access to GPU. Future work will explore Deep Learning architectures (LADDER, Deep Gaussian Processes) and budding Ef-KAN methods to further improve non-parametric cosmological inference.

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## Chapter 1

# Introduction

Over the past decade, the detection of gravitational waves (GWs) by the LIGO/Virgo Collaborations has revolutionized observational cosmology (Abbott et al., 2021). The landmark observations of GW150914 and GW170817 confirmed the existence of binary black hole and neutron star mergers, respectively, and demonstrated the power of GWs as new cosmic messengers. Building on this success, future missions such as the space-based Laser Interferometer Space Antenna (LISA) (Klein et al., 2016, Tamanini et al., 2016) and the ground-based Einstein Telescope (ET) (Sathyaprakash, Schutz, and Van Den Broeck, 2010, Maggiore et al., 2020) designed to extend GW observations to higher redshifts, opening a completely new window onto the expansion history of the Universe.

Despite remarkable progress by traditional approaches, a persistent *Hubble tension* remains: early-time measurements (using inverse distance ladder) from the Planck satellite ( $H_0 \approx 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ) differ by approximately  $5\sigma$  (Mukherjee et al., 2024) from late-time determinations (several low-redshift ( $z$ ) measurements) by the SH0ES team ( $H_0 \approx 73.3 \pm 1.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$ ). Direct measurements of  $H_0$  (Weinberg, 2008) traditionally rely on the distance–redshift relation of Cepheid-calibrated Type Ia supernovae observed with the Hubble Space Telescope (HST). Indirect methods include:

- **Type Ia Supernovae (SNe Ia)** as standard candles,
- **Cosmic Microwave Background (CMB)** angular scale measurements,
- **Baryon Acoustic Oscillations (BAO)** in galaxy clustering.

While these techniques achieve percent-level precision, they depend on multi-step calibrations and cosmological model assumptions, motivating the search for independent, calibration-free probes (Di Valentino et al., 2021).

Gravitational waves from compact-binary mergers act as *standard sirens*, providing direct measurements of the luminosity distance  $d_L(z)$  without intermediate distance ladders (Di Valentino et al., 2021). When combined with redshift information from electromagnetic counterparts or host-galaxy identification, standard sirens produce a model-independent measurement of the distance–redshift relation and therefore the Hubble parameter,

$$H(z) = \frac{c(1+z)^2}{(1+z)d_L'(z) - d_L(z)} \quad (1.1)$$

Accurate estimation of the Hubble parameter  $H(z)$  across cosmic time strengthens prediction of  $H_0 = H(z=0)$  which sets every scale in the Universe. Thereby, enables us to infer:

- Age of the Universe,
- Masses of clusters and galaxies,
- Energy budget of luminous sources,
- Physical scales of cosmic objects,
- Constraints on dark matter,

- Neutrino mass constraints,
- Constraints on  $\Lambda$ CDM extension parameter(s),
- Model independent cross-checks on the Hubble tension.

Future GW missions— the space-based **LISA**, sensitive to massive binary black hole (MBBH) mergers up to  $z \sim 8$ , and the 3rd generation ground-based **Einstein Telescope (ET)**, capable of detecting binary neutron star (BNS) mergers up to  $z \sim 2$ —are expected to detect numerous standard sirens. Their high event rates and extended redshift reach will significantly improve the precision of  $H(z)$  reconstructions.

Machine learning techniques, particularly *Gaussian Process Regression* (GPR), offer a flexible, non-parametric framework for reconstructing  $d_L(z)$  and its derivatives directly from noisy GW data. By leveraging the predictive power of GPs, one can recover a smooth estimate of  $H(z)$  without assuming a specific cosmological model, thus fully exploiting the model-independence of standard sirens.

In this work, I have generated realistic mock catalogs for LISA and ET, applied GPR to reconstruct the Hubble parameter over the redshift range  $0 \leq z \leq 4$  for LISA. I have done some tests to verify my GPR too. I have also explored the effect of Artificial Neural Networks (ANNs) when used for these reconstructions.



## Chapter 2

# Problem Statement & Literature Review

## 2.1 Problem Statement

Accurate reconstruction of the expansion history of the Universe (Hubble parameter) is a cornerstone of modern cosmology. Traditional probes of the Hubble parameter, including:

- **Type Ia Supernovae (SNe Ia)** as standard candles,
- **Cosmic Microwave Background (CMB)** angular scale measurements,
- **Baryon Acoustic Oscillations (BAO)** in galaxy clustering,

achieve impressive precision but require multi-step calibrations and assume a specific cosmological model ( $\Lambda$ CDM). These calibration procedures introduce systematic uncertainties and potential biases in the inferred  $H(z)$ .

Gravitational waves (GWs) from compact-binary mergers offer an independent alternative via the *standard siren* approach: direct measurement of the luminosity distance  $d_L(z)$  from the waveform, paired with host-galaxy redshift. However, current GW catalogs (LIGO/Virgo) are limited in number (tens of events) and confined to low redshift  $z \lesssim 0.1$ , preventing a decisive test of the Hubble tension between early-time (Planck) and late-time (SH0ES) determinations of  $H_0$ .

I therefore seek a complementary model-independent estimator of the expansion rate extending to high redshift. The central question of this dissertation is:

*Can future GW missions such as **LISA**—when combined with non-parametric machine learning techniques such as Gaussian Process Regression (GPR), accurately reconstruct the Hubble parameter  $H(z)$  at least up to  $z \sim 4$  (and that too using numerical derivatives) and thereby provide competitive constraints on the Hubble constant  $H_0$ ?*

## 2.2 Literature Review

The primary inspiration for this work is the paper “*Reconstructing the Hubble Parameter with Future Gravitational-Wave Missions Using Machine Learning*” by Mukherjee et al., 2024, which:

- Creates realistic mock catalogs for LISA and ET using early- and late-time cosmological priors.
- Applies Gaussian Process Regression (GPR) to reconstruct  $d_L(z)$ , its kernel derivatives, and hence  $H(z)$ .
- Demonstrates that a 10-year LISA run with  $\sim 80$  bright sirens can match a 3-year ET run with  $\sim 1000$  events in constraining  $H_0$ .
- Explores the dependence of reconstruction precision on mission duration, event rates, and priors.

In addition, foundational methodologies and forecasts include:

- **Gaussian Processes for Machine Learning:** Rasmussen & Williams (2006) provides the statistical underpinnings of Gaussian processes and kernel choices (Rasmussen and Williams, 2005).
- Forecasting  $F(Q)$  cosmology with CDM background using standard sirens by Ferreira et al., 2022, for suggestions to generate the mock catalogs.

Other works cited throughout this dissertation (see Bibliography) provide additional context on detector sensitivities, statistical inference techniques (MCMC, hyperparameter marginalization), and alternative machine learning approaches.

## Chapter 3

# Objectives & Hypothesis

### 3.1 Objectives

- Generate astrophysically realistic mock bright standard-siren catalogs for **LISA** ( $0 \leq z \leq 4$ ) and **ET** ( $0 \leq z \leq 2$ ), incorporating relevant sources of noise.
- Implement Gaussian Process Regression (GPR) for LISA to non-parametrically reconstruct the luminosity distance  $d_L(z)$ , its derivatives, and hence the Hubble parameter

$$H(z) = \frac{c(1+z)^2}{(1+z)d'_L(z) - d_L(z)} \quad (3.1)$$

- Validate the GPR reconstruction via cross-checks on noise-free mean catalogs and analytic test cases.
- Test different techniques to reduce errors due to computing derivatives of luminosity distance ( $d_L(z)$ ) numerically using `np.gradient`.
- Explore an Artificial Neural Network (ANN) approach to reconstruct at least the Hubble diagram, and compare its performance and uncertainties against the GPR method visually.

### 3.2 Hypothesis

*Given astrophysically realistic bright standard-siren events from future gravitational-wave missions (here, LISA), Gaussian Process Regression can robustly recover the expansion history  $H(z)$  to a good visual accuracy over the redshift range  $0 \leq z \leq 4$ . Consequently, this model-independent non-parametric approach will yield competitive constraints on the Hubble constant  $H_0$ , helping to address and potentially alleviate the existing tension between early- and late-time measurements.*

## Chapter 4

# Methodology

This chapter portrays the frameworks used in this dissertation to simulate gravitational-wave bright standard sirens mock data separately for ET & LISA and reconstruct the cosmic expansion history ( $H(z)$ ) for LISA. It comprises three main workflows: (A) Gaussian Process Regression (GPR) applied to generated realistic mock LISA catalogs, (B) generation of mock catalogs for the Einstein Telescope (ET), and (C) implementation of Artificial Neural Networks (ANNs) to LISA mock data to reconstruct the Hubble diagram at a higher redshift range.

### 4.1 Cosmological Framework

I have adopted Vanilla  $\Lambda$ CDM cosmology (Weinberg, 2008) with state-of-the-art fiducial parameters:

$$\begin{aligned} H_0 &= 70 \text{ km s}^{-1} \text{ Mpc}^{-1}, \\ \Omega_m &= 0.3, \\ \Omega_\Lambda &= 0.7. \end{aligned}$$

All distance calculations use

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}, \quad d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}.$$

#### Notation and Definitions

- $H_0$  Present-day Hubble constant, in units of  $\text{km s}^{-1} \text{ Mpc}^{-1}$
- $\Omega_m$  Matter density parameter (fraction of critical density in matter)
- $\Omega_\Lambda$  Dark energy density parameter (fraction of critical density in vacuum energy)
- $c$  Speed of light in vacuum (typically  $c = 2.998 \times 10^5 \text{ km s}^{-1}$ )
- $E(z)$  Dimensionless Hubble parameter,

$$E(z) = \frac{H(z)}{H_0} = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda},$$

- $d_L(z)$  Luminosity distance at redshift  $z$ , defined by

$$d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}.$$

### 4.2 Mock LISA Catalogs

Following Ferreira et al., 2022 and Tamanini et al., 2016, MBHB standard sirens have been simulated up to  $z \sim 9$ :

### 4.2.1 Weak Lensing and Peculiar Velocity Error Models

In simulating LISA distance measurements, I have taken account for two main sources of scatter in the inferred luminosity distance  $d_L(z)$ : weak gravitational lensing along the line of sight, and peculiar motions of the host galaxies. The parametrizations below mirror the implementation in code.

**Weak Lensing Dispersion** Small-scale density fluctuations induce magnification variance that grows with redshift. I thereby adopt the empirical form

$$\sigma_{\text{lens}}(z) = d_L(z) \times 0.066 \left[ \frac{1 - (1+z)^{-0.25}}{0.25} \right]^{1.8},$$

which ensures  $\sigma_{\text{lens}}(0) = 0$  and reproduces the  $z^{1.8}$ -scaling found in ray-tracing studies.

**Peculiar Velocity Uncertainty** Nearby hosts carry random line-of-sight motions, adding a Doppler uncertainty to  $d_L$ . I thereby model this as

$$\sigma_{\text{pv}}(z) = d_L(z) \frac{v_{\text{rms}}}{c}, \quad v_{\text{rms}} = 500 \text{ km s}^{-1},$$

so that at  $z \rightarrow 0$  the error vanishes and at higher  $z$  it decays as  $1/(1+z)$  in full Doppler treatments.

### 4.2.2 Redshift Data Generation

To populate the mock LISA catalog with  $N$  source redshifts, the following logical steps have been employed, inspired by astrophysical formation-delay scenarios:

**Scenario Selection** The user selects one of three astrophysical delay models—“Delay”, “Pop III” or “No Delay”. Each is assigned a pair of Beta-distribution shape parameters  $(\alpha, \beta)$ :

$$(\alpha, \beta) = \begin{cases} (2.42, 3.84), & \text{Delay} \\ (2.64, 6.03), & \text{Pop III} \\ (2.14, 4.70), & \text{No Delay.} \end{cases}$$

**Oversampling Loop** An oversized batch of redshifts has been repeatedly drawn from  $\text{Beta}(\alpha, \beta)$  scaled to  $[0, z_{\text{max}}]$ . To ensure efficiency under tight tails, each batch size is set to  $\lceil (N - N_{\text{kept}}) \times 2.5 \rceil + 50$ . I have accumulated samples until at least  $N$  fall within the valid window  $[z_{\text{min}}, z_{\text{max}}]$  or until a safe cap on total attempts is reached.

**Range Filtering** From each draw, retained only those  $z$ -values satisfying

$$z_{\text{min}} \leq z \leq z_{\text{max}},$$

discarding any outside the user-specified bounds.

**Convergence Check** If after the maximum allowed trials we still lack  $N$  valid points, a warning is issued indicating how many could be generated in  $[z_{\text{min}}, z_{\text{max}}]$ .

**Final Assembly** The first  $N$  accepted samples are:

1. Truncated to length  $N$ .
2. Passed through a uniqueness filter to remove duplicates.
3. Sorted in ascending order.
4. Packaged into a column vector for downstream use.

This method guarantees a representative redshift ensemble drawn from astrophysically motivated distributions, while controlling for boundary effects and statistical duplicates.

### 4.2.3 Simulating Observed Luminosity Distances for LISA

To emulate astrophysically realistic observational data for standard sirens, I perturb the theoretical luminosity distance  $d_L(z)$  using two independent uncertainty components—weak lensing and peculiar velocity effects—resulting in a noisy observable  $d_L^{\text{obs}}(z)$ . The logic behind the simulation is described as follows:

#### Input Handling

The redshift array  $z_i$  is accepted in 1D flattened form, and passed to the theoretical distance calculator  $d_L^{\text{theory}}(z)$  along with error functions.

#### Error Components

For each redshift  $z_i$ , let's evaluate:

- $\sigma_{\text{lens}}(z_i)$  — due to weak lensing magnification scatter.
- $\sigma_{\text{pv}}(z_i)$  — from peculiar motion of host galaxies.

Each is clipped to ensure non-negativity before proceeding.

#### Quadrature Combination

The total standard deviation for each point is obtained via:

$$\sigma_{\text{tot}}(z_i) = \sqrt{\sigma_{\text{lens}}^2 + \sigma_{\text{pv}}^2}.$$

#### Noise Floor Handling

To prevent unrealistically small uncertainties (especially at intermediate  $z$ ), a soft floor is imposed:

$$\sigma_{\text{tot}}(z_i) \geq \epsilon d_L(z_i), \quad \text{with } \epsilon \approx 10^{-5}.$$

A special rule is enforced for  $z = 0$ , where  $\sigma = 0$  and  $d_L = 0$  exactly, preserving the boundary condition.

### 4.2.4 Gaussian Sampling

For each redshift  $z_i$ , draw a random Gaussian fluctuation  $\delta_i \sim \mathcal{N}(0, \sigma_{\text{tot}})$  and define the observed value as:

$$d_L^{\text{obs}}(z_i) = d_L(z_i) + \delta_i.$$

#### Final Output

Two arrays are returned:

- $d_L^{\text{obs}}$  — simulated measurements with noise,
- $\sigma_{\text{tot}}$  — associated  $1\sigma$  uncertainties.

This methodology ensures a statistically robust generation of observationally realistic mock data, mimicking the systematics LISA will contend with, while safeguarding against numerical instabilities.

### 4.2.5 Enforcing the $d_L(0) = 0$ Constraint

To guarantee proper physical behavior and improve numerical stability in Gaussian Process Regression (GPR), the below condition has been explicitly imposed:

$$d_L(0) = 0,$$

which reflects the fact that the luminosity distance to a source at redshift zero must vanish.

**Purpose:** Incorporating the point ( $z = 0, d_L = 0$ ) ensures the GPR model is anchored at the origin, helping avoid unphysical extrapolations and improving the regression's fit near low redshift.

#### Implementation Steps

1. **Input Verification:** The function first checks whether  $z = 0$  is already included in the redshift data. If so, it modifies the corresponding  $d_L$  and error to enforce the constraint.
2. **Prepending Constraint:** If  $z = 0$  is absent, the following values are prepended to the dataset:
  - $z = 0$  is added to the redshift array,
  - $d_L = 0$  is added to the luminosity distance array,
  - A very small uncertainty value  $\sigma_0 = 10^{-9}$  is assigned to avoid exact zero, which could cause singularities in matrix inversions during GP fitting.
3. **Squared Error Calculation:** The errors are squared to obtain the variance array used as the  $\alpha$  parameter in GPR. A minimum floor of  $10^{-18}$  is enforced for the variance at  $z = 0$  to avoid computational instability.
4. **Output:** The modified arrays

$$(z, d_L^{\text{obs}}, \sigma_{d_L}, \sigma_{d_L}^2)$$

are returned, now with the origin constraint included.

This step, though seemingly minor, is crucial for ensuring the GPR model respects known boundary conditions and delivers physically plausible reconstructions of cosmological distances.

### 4.2.6 Initial Dataset Generation for GP Hyperparameter Tuning

To construct a reliable baseline for tuning the Gaussian Process (GP) kernel parameters, I generate a synthetic dataset that simulates gravitational-wave events with physically realistic redshift distributions and distance uncertainties. The steps involved in this process are summarized as follows:

#### Scenario Selection

A specific astrophysical delay model is chosen—in this case for example, the “No Delay” scenario—representing the distribution of mergers as a function of redshift. The redshifts are drawn using the logic outlined in above sections, with:

$$N = 80, \quad z \in [0.01, 9.0].$$

### Mock Data Generation

For each generated redshift  $z_i$ , the corresponding theoretical luminosity distance  $d_L(z_i)$  is perturbed with stochastic errors as described in Section 4.2.3. This yields the arrays:

- $d_L^{\text{obs}}(z_i)$  — simulated observed distances,
- $\sigma_{d_L}(z_i)$  —  $1\sigma$  observational uncertainties.

### Incorporation of the $z = 0$ Constraint

The  $(z = 0, d_L = 0)$  boundary condition is prepended to the dataset, with a minimal error assignment:

$$\sigma_{d_L}(0) = 10^{-9},$$

ensuring GP regularization while preserving physical correctness. The modified arrays

$$(z_i, d_L^{\text{obs}}(z_i), \sigma_{d_L}(z_i), \sigma_{d_L}^2(z_i))$$

now contain  $N + 1$  entries, suitable for hyperparameter optimization.

### Outcome

This curated dataset forms the basis for subsequent kernel tuning using Bayesian optimization and MCMC. It captures the full complexity of redshift-dependent uncertainty while remaining computationally efficient for repeated GP evaluations.

### 4.2.7 Visualizing the Initial Dataset

To gain insight into the quality and distribution of the simulated luminosity distance data, visualization of the initial dataset used for hyperparameter tuning against the theoretical prediction from the fiducial  $\Lambda$ CDM model is added.

**Theoretical Reference Curve** We compute the theoretical luminosity distance  $d_L(z)$  on a dense grid over the redshift interval  $z \in [0, 9]$ , using the fiducial cosmology parameters:

$$H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad \Omega_m = 0.3, \quad \Omega_\Lambda = 0.7.$$

This curve, rendered as a dashed black line, serves as the baseline for evaluating the fidelity of the simulated data.

**Simulated Data Points** Each point in the mock dataset, including the enforced origin  $(0, 0)$ , is plotted with vertical error bars (See Results). These bars reflect the total observational uncertainty  $\sigma_{d_L}$  combining weak lensing and peculiar velocity effects (see Sections 4.2 and 4.2.3).

The number of events simulated is  $N + 1 = 81$  in this case, with the redshift values spanning  $z \in [0.01, 9.0]$  and one added point at  $z = 0$ .

**Interpretation:** This visualization (See Results) verifies that:

- The sampled redshifts cover the intended range.
- The scatter around the theoretical curve is consistent with the modeled uncertainties.
- The  $z = 0$  anchor point behaves as expected and helps constrain the GP interpolation near the origin.

Overall, this figure provides an important sanity check before proceeding to hyperparameter tuning and model training.



## 4.3 Gaussian Process Regression for LISA

### 4.3.1 GP Formalism

I start with a brief overview of Gaussian Process Regression (GPR), a non-parametric machine learning method used to model distributions over functions directly in a continuous domain (Rasmussen and Williams, 2005, Seikel and Clarkson, 2013, Shafieloo, Kim, and Linder, 2012). At its core, a Gaussian Process (GP) is a collection of random variables, any finite number of which follow a joint multivariate normal (MVN) distribution. This framework allows us to reconstruct a smooth function from data without assuming any specific parametric form.

In the context of this work, I apply GPR to reconstruct the luminosity distance  $d_L(z)$  from a mock catalog of redshift–luminosity distance pairs. The key assumption is that both the noisy data and the true underlying function are jointly Gaussian-distributed, fully described by a mean function and a covariance kernel.

Specifically, for any redshift  $z$ , the predicted mean and uncertainty of the function are given by:

$$\mu(z) = \mathbb{E}[d_L(z)], \quad \text{Var}[d_L(z)] = \kappa(z, z),$$

where  $\kappa(z, \tilde{z})$  is the covariance function (kernel) encoding correlations between points at redshifts  $z$  and  $\tilde{z}$ . I adopt a **zero mean** prior to ensure that the reconstruction remains agnostic to cosmological models. Therefore, the choice of kernel entirely governs the behavior and smoothness of the reconstructed function.

I thereby model the joint distribution of observed distances and the underlying function  $d_L(z)$  as a multivariate normal with zero mean and Matérn  $\nu = 9/2$  kernel:

$$\kappa(z, z') = \sigma_f^2 \exp\left(-\frac{3|z - z'|}{\ell}\right) \left[1 + \frac{3|z - z'|}{\ell} + \frac{27(z - z')^2}{7\ell^2} + \frac{18|z - z'|^3}{7\ell^3} + \frac{27(z - z')^4}{35\ell^4}\right],$$

where  $(\sigma_f, \ell)$  are the characteristic variance and length scale.

### Hyperparameter Inference

I sample the posterior  $p(\sigma_f, \ell \mid \{z_i, d_L^{\text{obs}}, \sigma_{d_L}\})$  via the `emcee` ensemble MCMC sampler, marginalizing the log-marginal likelihood rather than point-estimating.

### Reconstruction and Derivatives

Given posterior samples of  $(\sigma_f, \ell)$ , we compute the predictive mean and covariance of  $d_L(z)$  at test redshifts. Numerical derivatives  $d'_L(z)$  are obtained using central finite differences (`np.gradient`). The Hubble parameter is then

$$H(z) = \frac{c(1+z)^2}{(1+z)d'_L(z) - d_L(z)} \quad (4.1)$$

and uncertainties on  $H(z)$  and  $H_0$  are also propagated.

### 4.3.2 Log-Posterior for GP Hyperparameter Optimization

To identify optimal kernel hyperparameters for Gaussian Process Regression (GPR), I define a log-posterior function that combines data-driven likelihood with soft prior bounds. This function is used by both Bayesian Optimization and Markov Chain Monte Carlo (MCMC) algorithms to guide the search.

**Parameterization:** The two key hyperparameters are:

- $\ell$ : the correlation length (or smoothness scale),
- $\sigma_f$ : the signal variance (amplitude scale).

We perform optimization in log-space using:

$$\log \ell, \quad \log \sigma_f.$$

**Priors:** I impose weak, uniform priors on the log-parameters within physically reasonable bounds:

$$\begin{aligned} \log \ell &\in [\log 0.1, \log 25.0], \\ \log \sigma_f &\in \left[ \log \left( \frac{\hat{\sigma}_{\text{data}}}{100} \right), \log(100 \hat{\sigma}_{\text{data}}) \right], \end{aligned}$$

where  $\hat{\sigma}_{\text{data}}$  is estimated as the standard deviation of  $d_L^{\text{obs}}$ , excluding the  $z = 0$  anchor point. These constraints avoid degenerate kernel shapes or numerical instabilities.

**GP Kernel:** We adopt a smooth Matérn kernel with  $\nu = 4.5$ , equivalent to differentiability up to fourth order ( $C^4$ ). This choice ensures that the GP model supports smooth derivatives, which is necessary for calculating  $H(z)$ :

$$k(z_i, z_j) = \sigma_f^2 \mathcal{M}_{4.5}(|z_i - z_j|; \ell).$$

**Likelihood Computation:** Given input data  $\{z_i, d_L^{\text{obs}}(z_i)\}$  and their noise variances  $\sigma_i^2$ , the GP regressor is constructed with  $\alpha = \sigma_i^2$ . The log-marginal likelihood  $\log \mathcal{L}_{\text{GP}}$  is evaluated directly after fitting the GP model. If this operation fails due to ill-conditioning or yields a NaN, the function returns  $-\infty$  to penalize the sampled hyperparameter pair.

**Final Log-Posterior:** Since the priors are flat (uniform in log-space), the log-posterior simplifies to:

$$\log \mathcal{P}(\theta | \mathcal{D}) = \log \mathcal{L}_{\text{GP}}(\theta) + \text{const.},$$

with  $\theta = (\log \ell, \log \sigma_f)$ .

This formulation enables robust and efficient hyperparameter inference while controlling for overfitting and degeneracies.

### 4.3.3 Bayesian Optimization for Initial Hyperparameter Selection

To provide a strong initialization for further hyperparameter tuning (e.g., via MCMC), I employ Bayesian Optimization (BO) to find a good pair of kernel parameters  $(\ell, \sigma_f)$  for the Gaussian Process Regression (GPR) model.

**Search Space Construction:** The search domain is defined in real-space but scaled logarithmically to accommodate the natural variability of the parameters:

$$\begin{aligned} \ell &\in [0.1, 25.0], \\ \sigma_f &\in \left[ \max \left( \frac{\hat{\sigma}_{\text{data}}}{100}, 10^{-2} \right), 100 \hat{\sigma}_{\text{data}} \right], \end{aligned}$$

where  $\hat{\sigma}_{\text{data}}$  is the standard deviation of the observed luminosity distances (excluding the  $z = 0$  point). This ensures that the signal amplitude is aligned with the data's intrinsic scale.

**Objective Function:** The BO framework minimizes the negative log-posterior defined in Section 4.3.2:

$$\text{Objective: } \min_{(\ell, \sigma_f)} \left[ -\log \mathcal{P}(\ell, \sigma_f | \mathcal{D}) \right],$$

subject to the constraint that failed evaluations return a large finite penalty (e.g.,  $10^{18}$ ) rather than infinity, to maintain numerical stability during optimization.

### Optimization Settings

- **Initial Samples:** 15 random evaluations to explore the space.
- **BO Iterations:** 60 total calls to refine the minimum.
- **Acquisition Function:** Expected Improvement (EI), which balances exploitation of known good regions with exploration of uncertain areas.

The entire process is timed to assess computational efficiency.

**Expected Results** After 60 calls, the BO algorithm reports the best-fit hyperparameters (See Results) which are used as the starting point for more exhaustive sampling in the MCMC stage.

**Convergence Plot** The convergence history of the BO iterations is also visualized (See Results), showing how the minimum value of the negative log-posterior improves over successive evaluations.

This process ensures that the kernel hyperparameters are initialized in a region of parameter space that is both physically meaningful and numerically stable, substantially improving downstream performance and convergence.

### 4.3.4 Refining GP Hyperparameters Using MCMC

Following Bayesian Optimization (BO), Markov Chain Monte Carlo (MCMC) is applied to refine the estimates of the Gaussian Process (GP) kernel hyperparameters: correlation length  $\ell$  and signal amplitude  $\sigma_f$ . This allows for a more accurate sampling of the posterior distribution and provides uncertainty estimates.

**Posterior Analysis** After discarding the burn-in samples, the remaining chain is flattened and used to:

- Compute marginal posterior distributions of  $\ell$  and  $\sigma_f$ ,
- Extract the medians and 68% credible intervals:

$$\ell_{\text{MCMC}} = \text{median}_{\text{chain}}(\exp(\theta_0)), \quad \sigma_{f,\text{MCMC}} = \text{median}_{\text{chain}}(\exp(\theta_1)),$$

- Visualize (See Figure) the 2D posterior using corner plots (if supported).

**Final Output** The refined values of  $\ell$  and  $\sigma_f$  are used as fixed hyperparameters for all subsequent GP reconstructions:

$$\text{Fixed: } \ell = \ell_{\text{final}}, \quad \sigma_f = \sigma_{f,\text{final}},$$

where these are either the MCMC medians or the BO best guesses, depending on the success of the MCMC procedure.

This refinement phase allows for a fully probabilistic characterization of the GP kernel, capturing the epistemic uncertainty in the data-driven interpolation model.

### 4.3.5 Main Simulation Loop

To quantify the uncertainty and stability in the reconstruction of the Hubble parameter  $H(z)$  using Gaussian Process Regression (GPR), I perform a Monte Carlo ensemble of  $N_{\text{sim}}$  synthetic simulations. Each simulation involves the full computational pipeline of redshift sampling, data generation, GP regression, and derivative-based cosmological inference.

**Prediction Grid** All predictions are evaluated on a dense redshift grid:

$$z \in [0, z_{\max}], \quad \text{with} \quad N_{\text{grid}} = 100,001,$$

ensuring high resolution for numerical derivatives and interpolation smoothness.

**Fixed Kernel** The same GPR kernel is used across all simulations. It is constructed from the MCMC-refined or BO-derived hyperparameters:

$$\kappa(z_i, z_j) = \sigma_f^2 \mathcal{M}_{4.5}(|z_i - z_j|; \ell),$$

with  $\nu = 4.5$  ensuring sufficient smoothness for differentiability up to fourth order.

**Simulation Workflow** Each simulation proceeds as follows:

1. **Redshift Sampling:** Draw  $N$  redshifts from the specified astrophysical scenario (e.g., No Delay), ensuring they lie within the bounds  $z \in [0.01, 9.0]$ .
2. **Mock Data Generation:** For each sampled redshift, generate observed  $d_L^{\text{obs}}(z)$  values with combined weak lensing and peculiar velocity errors.
3. **Apply  $z = 0$  Constraint:** Append  $(z = 0, d_L = 0)$  with a small error to ensure correct boundary behavior.
4. **GP Regression:** Fit a non-optimized GPR model using the fixed kernel and simulation-specific noise variances.
5. **Prediction:** Use the fitted GP to predict  $d_L(z)$  over the entire redshift grid.
6. **Derivative Estimation:** Compute the numerical gradient of  $d_L(z)$  via central differences to obtain  $d'_L(z)$ .
7. **Hubble Parameter Calculation:** Reconstruct the Hubble parameter using:

$$H(z) = \frac{c(1+z)^2}{d'_L(z)(1+z) - d_L(z)}$$

applied only at points where the denominator is physically valid (i.e.,  $> 0.1$ ). Special treatment is applied at  $z = 0$ , using  $H(0) = c/d'_L(0)$  if the derivative is well-defined.

8. **Result Storage:** The predicted  $d_L(z)$ ,  $d'_L(z)$ , and  $H(z)$  for each simulation are stored for ensemble statistics.

**Monitoring and Success Criteria:** Progress updates are printed every 10 simulations. Simulations that fail due to GP fitting errors or unstable numerics are skipped. A final tally of successful simulations and total runtime is reported:

$$N_{\text{success}} \leq N_{\text{sim}}, \quad \text{Runtime: } \Delta t.$$

**Post-Simulation Use:** The aggregated results are later (See Results) used to:

- Estimate mean and standard deviation of  $H(z)$  across simulations,
- Evaluate ensemble uncertainties in  $d_L(z)$  and its gradient,
- Plot confidence bands and compare to theoretical cosmology.

This loop forms the backbone of the reconstruction methodology, allowing robust statistical conclusions about the capability of LISA to constrain the cosmic expansion history via standard sirens.

### Cross-Checks

**Noise-Free Validation:** Apply GPR to mock catalogs with  $\sigma_{d_L} = 0$  to confirm recovery of input  $d_L(z)$  (see Figure 5.7).

## 4.4 Mock Data Generation for ET

### 4.4.1 Cosmological Framework for Einstein Telescope (ET)

The Einstein Telescope (ET) simulations adopt the same cosmological background as the LISA simulations, based on a fiducial flat  $\Lambda$ CDM model with the following parameters:

$$\begin{aligned} H_0 &= 70 \text{ km s}^{-1} \text{ Mpc}^{-1} && \text{(Hubble constant),} \\ \Omega_m &= 0.3 && \text{(matter density parameter),} \\ \Omega_\Lambda &= 0.7 && \text{(dark energy density parameter),} \\ c &= 2.998 \times 10^5 \text{ km s}^{-1} && \text{(speed of light).} \end{aligned}$$

**Redshift Range for ET:** The ET instrument is sensitive primarily to low and intermediate redshifts. Accordingly, all simulations for ET standard sirens are restricted to the range:

$$z \in [0.07, 2.0].$$

**Dimensionless Expansion Function:** I define the normalized Hubble expansion function as:

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}.$$

**Hubble Parameter** The redshift-dependent Hubble parameter is reconstructed as:

$$H(z) = H_0 \cdot E(z),$$

with  $H_0$  in units of  $\text{km s}^{-1} \text{ Mpc}^{-1}$ .

**Luminosity Distance** The electromagnetic luminosity distance  $d_L(z)$ , in megaparsecs, is computed from:

$$d_L(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E(z')}.$$

This integral is evaluated numerically using either adaptive quadrature (for scalar inputs) or batch integration (for arrays of redshift values).

**Reference Grid** For visualization and theoretical comparisons,  $d_L(z)$  is precomputed on a dense redshift grid:

$$z_{\text{grid}} = \text{linspace}(0, 2.0, 1000),$$

yielding a smooth theoretical curve for the ET band.

These functions form the cosmological backbone for all synthetic bright standard siren datasets modeled for ET, supporting both forward modeling of observables and inverse reconstruction tasks.

### 4.4.2 Error Modeling and Mock Catalogue Generation for ET

To simulate observed luminosity distances for the Einstein Telescope (ET), I incorporate three key sources of measurement uncertainty into each mock event. These include instrument noise, gravitational lensing scatter, and a residual systematic term. The final observable is computed as:

$$d_L^{\text{obs}}(z) = d_L^{\text{theory}}(z) + \mathcal{N}(0, \sigma_{\text{tot}}(z)),$$

where  $\sigma_{\text{tot}}$  is the total uncertainty at redshift  $z$ .

**1. Instrumental Uncertainty:** The ET instrumental error is modeled using a polynomial expression in redshift, fit from sensitivity curves:

$$\sigma_{\text{inst}}^{\text{ET}}(z) = \left(0.1449z - 0.0118z^2 + 0.0012z^3\right) d_L^{\text{theory}}(z).$$

This captures redshift-dependent noise degradation in signal extraction.

**2. Weak Lensing Error:** Magnification fluctuations due to intervening large-scale structure introduce additional scatter:

$$\sigma_{\text{lens}}^{\text{ET}}(z) = 0.05z d_L^{\text{theory}}(z).$$

This linear scaling reflects increased line-of-sight structure at higher redshift.

**3. Systematic Error:** A fixed fractional uncertainty is added to reflect calibration, modeling, or population synthesis uncertainties:

$$\sigma_{\text{sys}}^{\text{ET}}(z) = \varepsilon d_L^{\text{theory}}(z), \quad \text{with } \varepsilon = 0.02.$$

**4. Total Uncertainty.** All components are combined in quadrature to obtain the total uncertainty:

$$\sigma_{\text{tot}}(z) = \sqrt{\sigma_{\text{inst}}^2(z) + \sigma_{\text{lens}}^2(z) + \sigma_{\text{sys}}^2(z)}.$$

**5. Mock Catalogue Construction:** Given an array of redshifts  $\{z_i\}$  drawn from the ET source distribution, I generate:

- $d_L^{\text{obs}}(z_i)$  — perturbed with Gaussian noise of width  $\sigma_{\text{tot}}(z_i)$ ,
- $\sigma_{\text{tot}}(z_i)$  — stored for uncertainty propagation and GP training.

**Validation** If the redshift array is empty, the generation step is skipped. Otherwise, the simulated data are ready for use in regression and reconstruction procedures.

This process yields a statistically consistent mock catalogue of ET observations (see Figure 5.1) that includes realistic and redshift-dependent sources of measurement error.

## Chapter 5

# Results

In this chapter I portray the reconstructed expansion history and Hubble constant constraints derived from the generated mock standard-siren catalogs. I begin with showing the plots of generated initial mock data for ET and LISA. I proceed to frame the visualizations (for LISA) of the GP reconstructed Hubble diagram, reconstructed  $d_L'$  vs  $z$  plot, and finally the graph of the reconstructed Hubble parameter. I also show the plot of ANN reconstructed  $d_L$  vs  $z$  and discuss the inferred observations from these plots.

### 5.1 Einstein Telescope (ET) Mock Catalogue Visualization

To verify the quality and distribution of the simulated Einstein Telescope (ET) standard siren data, visual of the observed luminosity distance measurements  $d_L^{\text{obs}}(z)$  alongside the fiducial theoretical curve  $d_L^{\Lambda\text{CDM}}(z)$  has been shown in Figure 5.1.

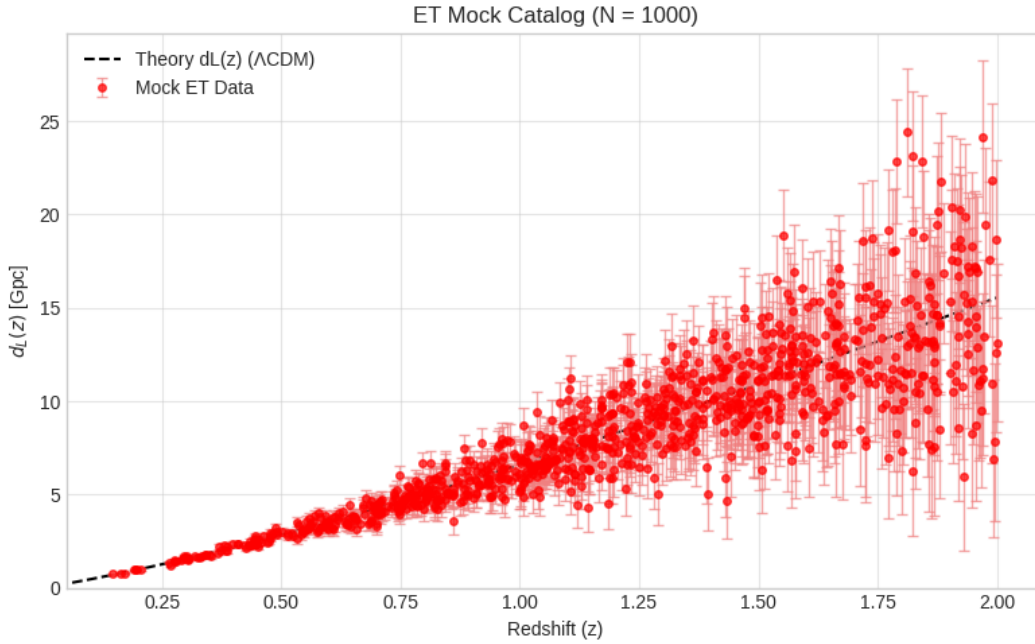


FIGURE 5.1: Mock ET luminosity distance data (red points with error bars) overlaid on the theoretical vanilla  $\Lambda\text{CDM}$  curve (dashed black line). The dataset contains  $N = N_{\text{events\_ET}} = 1000$  events with realistic instrumental, lensing, and systematic errors included.

#### 5.1.1 Plot Components

- The **theoretical curve**  $d_L^{\text{theory}}(z)$  is plotted as a dashed black line, computed using the fiducial cosmology described in Section 4.4.1.

- The **mock data points**  $(z_i, d_L^{\text{obs}}(z_i))$  are shown in red, with vertical error bars representing the total uncertainty  $\sigma_{\text{tot}}(z_i)$  from all ET noise sources (see Section 4.4.2).

### 5.1.2 Interpretation

The mock data points exhibit a characteristic scatter around the vanilla  $\Lambda$ CDM curve due to redshift-dependent observational noise. At low redshift, the errors are dominated by peculiar velocities, while at higher  $z$ , lensing and systematics dominate. The consistent alignment with the theoretical trend confirms the physical and statistical integrity of the simulated ET catalogue.

### 5.1.3 Axis Configuration

- The x-axis spans the ET redshift range  $z \in [0.07, 2.0]$ , with a small buffer for edge visibility.
- The y-axis begins at zero and extends based on the highest observed value of  $d_L(z)$ , converted from Mpc to Gpc.
- A grid and legend are included to enhance interpretability.

This figure serves as a critical validation step prior to applying regression models to the ET mock catalogue. I plan to apply GPR and other techniques in future to reconstruct the hubble diagram for ET and hence, derive the Hubble parameter reconstruction.

## 5.2 Initial Dataset (LISA) Visualization for Hyperparameter Tuning

To inspect the properties of the training dataset used for Gaussian Process Regression (GPR) hyperparameter tuning, we visualize the initial simulated catalog of luminosity distances  $d_L(z)$ , including the explicitly imposed anchor point at  $z = 0$ .

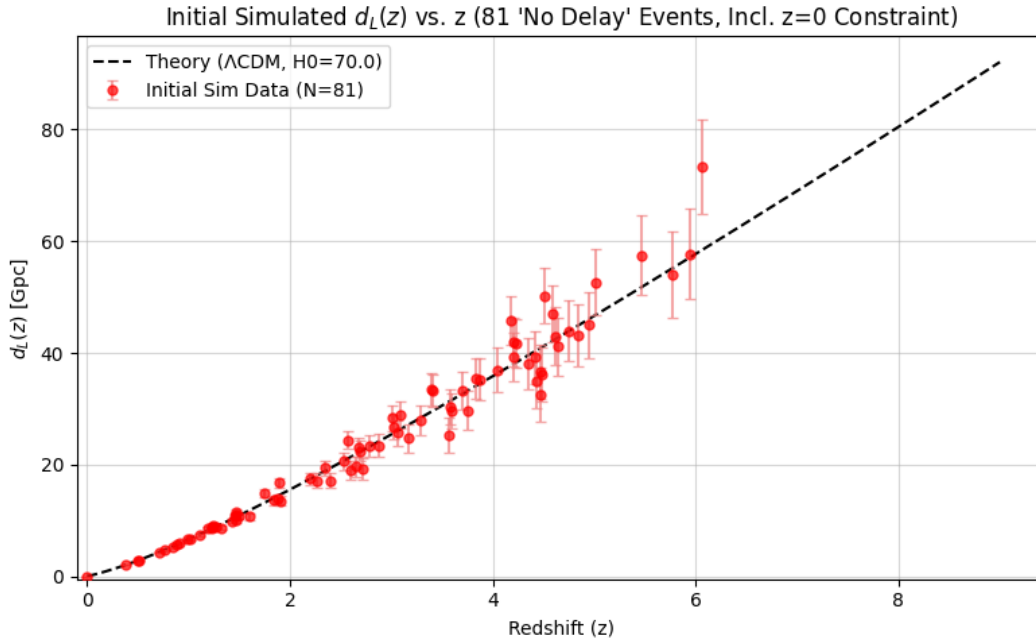


FIGURE 5.2: Initial simulated dataset (No Delay) used for GP hyperparameter tuning. The red points show mock data including observational noise and the  $z = 0$  constraint. The black dashed line represents the theoretical  $\Lambda$ CDM prediction for  $d_L(z)$ .



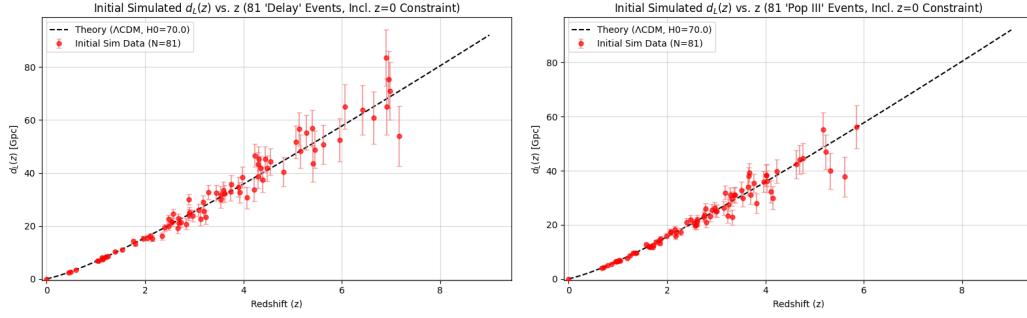


FIGURE 5.3: (Left) LISA mock data for "Delay" scenario; (Right) LISA mock data for "Pop III" scenario.

### 5.2.1 Plot Description

- The theoretical luminosity distance curve is shown (see Figure 5.2) as a black dashed line, computed over the redshift range  $z \in [0, 9]$ .
- The mock dataset includes  $N = \text{len}(z\_data\_initial)$  data points sampled from the chosen astrophysical delay scenario (here, "No Delay"), including a constraint at  $z = 0$  with  $d_L(0) = 0$ .
- Error bars represent the total uncertainty  $\sigma_{d_L}(z)$  combining weak lensing, and peculiar velocity contributions.

### 5.2.2 Interpretation

The red points cluster around the theoretical trend, as expected, with increasing spread at higher redshifts due to larger cumulative uncertainties. The addition of the  $z = 0$  point ensures that the model is properly constrained at the origin, which is critical for stable GPR training and reliable extrapolation near  $z = 0$ .

### 5.2.3 Axis Configuration

- The x-axis starts slightly below zero to clearly display the origin point.
- The y-axis is scaled based on the observed data range to fully capture all uncertainty bands.

This visualization confirms the statistical integrity and physical plausibility of the training data used for initial GPR kernel tuning. Plots obtained for "Delay" and "Pop III" scenarios have been shown in Figure 5.3.

## 5.3 GP Reconstructions for LISA

In this section I present the ensemble-averaged reconstructions of the luminosity distance  $d_L(z)$ , its derivative  $d'_L(z)$ , and the Hubble parameter  $H(z)$  obtained from  $N_{\text{sim}} = 100$  mock catalogues using Gaussian Process Regression (GPR). Each panel shows the mean reconstruction with  $1\sigma$ ,  $2\sigma$  and  $3\sigma$  confidence bands, compared to the fiducial  $\Lambda\text{CDM}$  prediction.

### 5.3.1 Reconstructed Luminosity Distance $d_L(z)$

The reconstructed  $d_L(z)$  (see Figure 5.4) closely tracks the theoretical curve out to  $z \approx 9$ . The one-sigma envelope encloses the majority of mock realizations, demonstrating that the GPR uncertainty (empirical spread) faithfully captures the stochastic noise injected by weak lensing and peculiar velocities. The inclusion of the  $z = 0$  constraint ensures that the model is anchored correctly at the origin.

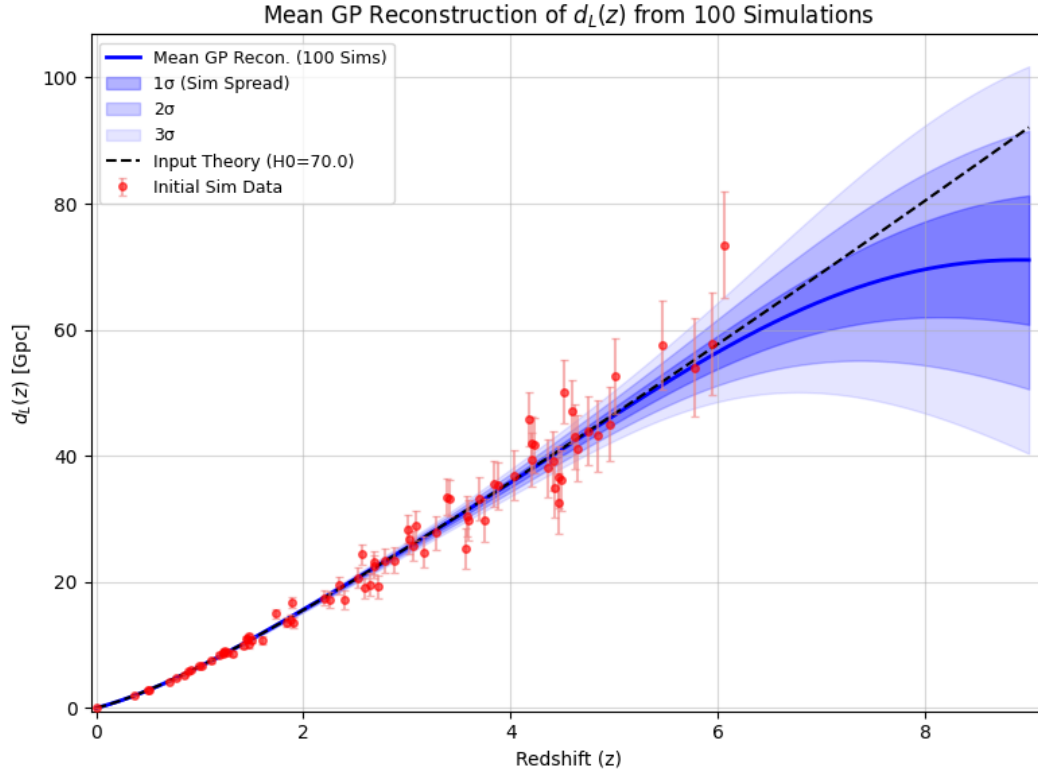


FIGURE 5.4: Mean GPR reconstruction of  $d_L(z)$  (blue solid line) with shaded bands indicating  $1\sigma$  (dark),  $2\sigma$  and  $3\sigma$  (lighter) spread across simulations. The red points and error bars show the initial mock data (including the  $z = 0$  anchor). The dashed black curve is the theoretical  $\Lambda$ CDM prediction.

### 5.3.2 Reconstructed $d'_L(z)$

The derivative  $d'_L(z)$  (see Figure 5.5) is estimated via finite differences on the GPR mean. Agreement with the theoretical slope is excellent at low to intermediate redshifts ( $z \lesssim 5$ ), with increasing scatter at higher  $z$  due to sparser sampling and larger observational noise. This derivative is a key ingredient for reconstructing the Hubble parameter.

### 5.3.3 Hubble Parameter Reconstruction $H(z)$

The full-range reconstruction of  $H(z)$  demonstrates that the GPR pipeline recovers the overall expansion history up to  $z \simeq 9$ , with the mean curve lying within the one-sigma band around the fiducial model. At high redshifts the uncertainty widens, reflecting the compounding effects of larger distance errors and numerical derivative noise.

Focusing on  $0 \leq z \leq 4$  (see Figure 5.6), where mock data density is higher and uncertainties lower, the GPR mean and one-sigma band tightly bracket the input  $H(z)$ . This zoomed view highlights the expected percent-level precision that future LISA standard sirens could achieve for the low-to-mid redshift.

#### Summary of Key Findings:

- The GPR ensemble faithfully reconstructs  $d_L(z)$  and its derivative across the full redshift range.
- The reconstructed  $H(z)$  agrees with the fiducial model within the empirical uncertainty bands, with relative errors growing at the highest redshifts.
- The  $z = 0$  constraint effectively anchors the low- $z$  behavior, improving stability in the derivative and  $H(0)$  estimation.

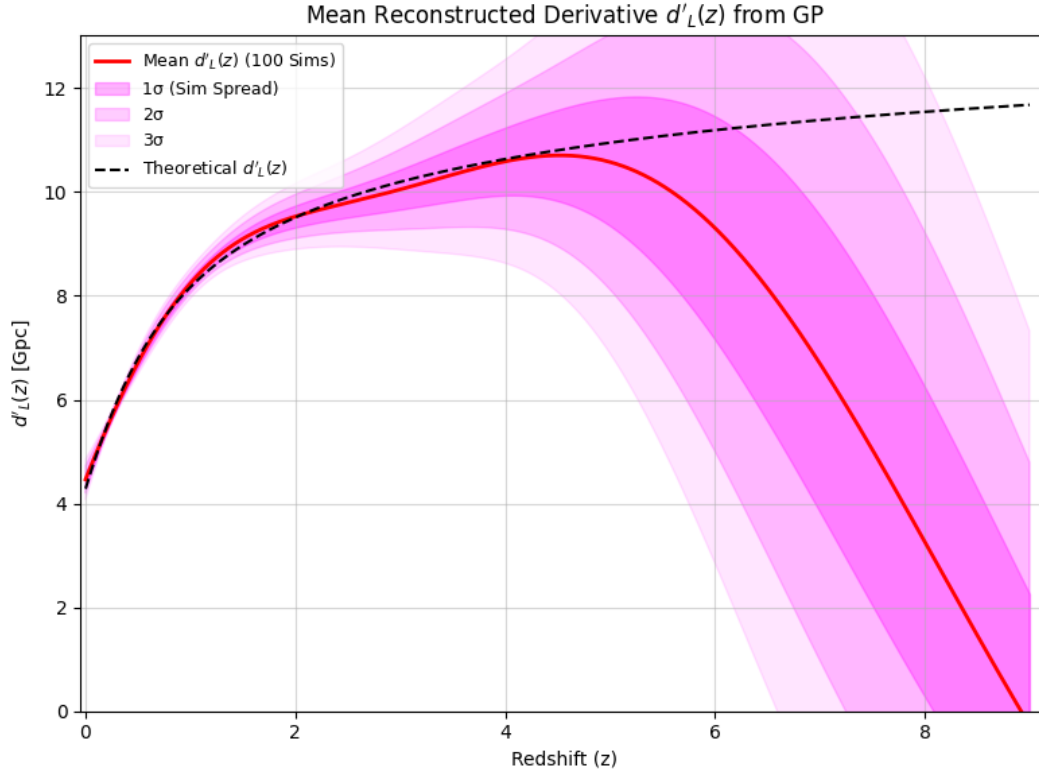


FIGURE 5.5: Mean reconstructed derivative  $d'_L(z)$  (red solid line) with  $1\sigma$ ,  $2\sigma$ ,  $3\sigma$  uncertainty bands. Dashed black line shows the theoretical derivative obtained by differentiating the fiducial  $d_L(z)$  curve.

- Denser the grid, smoother the derivatives.
- Mean reconstructed  $H_0 = 67.20$  km/s/Mpc, standard deviation of reconstructed  $H_0 = 1.91$  km/s/Mpc (from 100 simulations), relative uncertainty = 2.84 percentage, and deviation from fiducial = 2.80 ( $1.5\sigma$ ).

### Cross-Checks

**Noise-Free Validation:** I have applied GPR to mock catalogs with noise-free mean to verify the GP (see Figure 5.7).

## 5.4 Pros and Cons of Using Gaussian Process Regression (GPR) for Reconstructing $H(z)$

### 5.4.1 Pros (Advantages)

#### 1. Model-Independent

- GPR does **not assume** any specific cosmological model or form for  $H(z)$  — the data shape the result.
- This makes it especially useful for tension studies, like the Hubble tension, where we aim to avoid prior biases.

#### 2. Built-in Uncertainty Quantification

- GPR **naturally provides uncertainty estimates** at every point — not only predictions, but also confidence intervals.

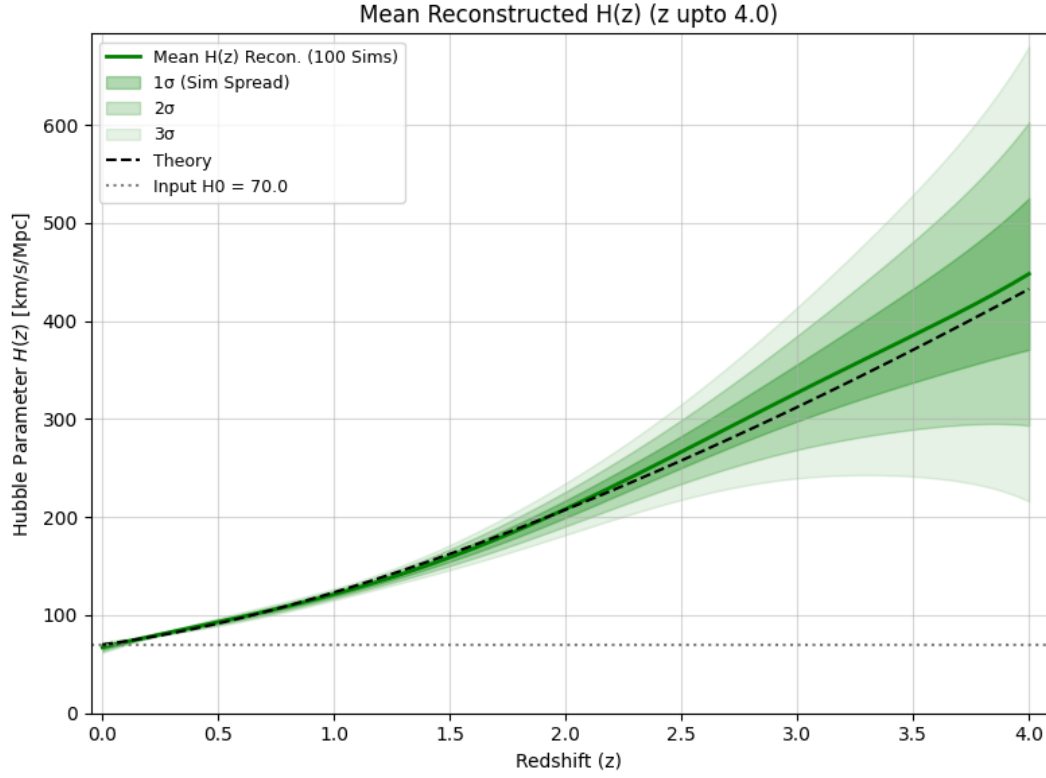


FIGURE 5.6: Zoomed view of the reconstructed  $H(z)$  for  $z \in [0, 4]$ . Mean reconstructed Hubble parameter  $H(z)$  (green solid line) with shaded uncertainties.

- This is crucial when reconstructing something like  $H(z)$  from noisy GW data.

### 3. Smooth Function Interpolation

- GPR ensures a **smooth reconstruction** — no sharp jumps, making derivatives (such as  $dL'(z)$ ) feasible.

### 4. Flexible & Adaptable

- Different kernel choices (like Matérn kernels) allow tuning smoothness and flexibility.
- We can adapt the kernel or hyperparameters depending on data quality.

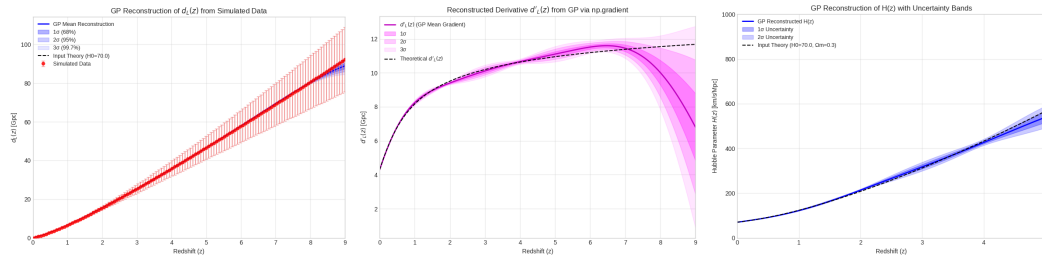


FIGURE 5.7: Verification of GPR for LISA using noise-free mean. Reconstructed  $H_0 = 69.70 \pm 0.07$  km/s/Mpc ( $1\sigma$ ).

## 5. Proven in Cosmology

- GPR has been successfully applied in other cosmological reconstructions, such as dark energy equation of state  $w(z)$ , Hubble parameter from SNe Ia, and cosmic chronometers.

### 5.4.2 Cons (Limitations)

#### 1. Computationally Expensive

- GPR requires **inverting an**  $N \times N$  covariance matrix (where  $N$  is the number of data points).
- The time complexity is  $O(N^3)$ , which can be heavy for large datasets (such as thousands of GW events in case of ET, DECIGO and other future GW missions).

#### 2. Derivative Amplifies Noise

- GPR reconstructs the function (such as  $d_L$ ), but **numerically differentiating** it to obtain  $H(z)$  introduces **additional noise**, especially at high redshifts where data is sparse.

#### 3. Kernel Choice Matters

- A poor choice of kernel can lead to either **over-smoothing** (missing real features) or **overfitting** (capturing noise).
- This requires careful hyperparameter tuning, which was handled using Bayesian Optimization and MCMC.

#### 4. Boundary Effects

- Near the edges of the data (e.g., near  $z = 0$  or at high  $z$ ), GPR reconstructions can become less reliable.
- This was mitigated partially by adding the constraint ( $z = 0, d_L = 0$ ).

#### 5. Assumes Gaussian Errors

- GPR technically assumes that the data noise is **Gaussian-distributed**. If real errors are strongly non-Gaussian, the results might be slightly biased.

While GPR gives model-independent, uncertainty-aware reconstructions, it does suffer from derivative noise amplification and scaling challenges for very large datasets. Hence, I have tried to find alternatives to GPR. I have found deep learning algorithms to be promising.

## Chapter 6

# Artificial Neural Network (ANN) Reconstruction of Hubble-Diagram (LISA)

In addition to Gaussian Process Regression (GPR), I have explored a deep learning approach using Artificial Neural Networks (ANNs) to reconstruct the luminosity distance function  $d_L(z)$  from simulated LISA standard siren data. This model was implemented using Keras with TensorFlow backend and incorporated Monte Carlo (MC) Dropout for regularization.

**Note on Derivatives** For this project, I focussed solely on reconstructing  $d_L(z)$  and not its derivatives, since numerical differentiation of ANN outputs was observed to produce significant instability compared to GPR. Consequently, no  $z = 0$  anchor point was enforced in the training set, and the model was used only to produce a Hubble diagram over  $z \in [0.01, 9.0]$ . Improving ANN-based derivative estimation remains an open direction for future work.

**Data Scaling** To ensure numerical stability during training, both input and output were standardized:

- Redshifts  $z$  were scaled using `StandardScaler`,
- Observed luminosity distances  $d_L^{\text{obs}}$  were similarly normalized.

### 6.1 Network Architecture

The model was constructed as follows:

- Input layer for one-dimensional  $z$  values,
- Two fully connected hidden layers with ReLU activation:
  - 64 units in the first hidden layer,
  - 32 units in the second hidden layer,
- Dropout layers with rate  $p = 0.1$  placed after each dense layer,
- L2 regularization ( $\lambda = 0.001$ ) applied to all dense layers,
- Output layer with a single linear neuron to predict  $d_L(z)$ .

**Training Procedure** The model was compiled using:

- **Loss:** Mean Squared Error (MSE),
- **Optimizer:** Adam with learning rate  $\eta = 10^{-3}$ ,
- **Metric:** Mean Absolute Error (MAE).

Training was performed over a maximum of 500 epochs using:

- Batch size of 16,
- 20% of the training data reserved for validation,
- **Early stopping:** Training stopped if validation loss failed to improve for 50 consecutive epochs, with best weights restored.

## 6.2 Results

To assess the predictive power of the Artificial Neural Network (ANN) with Monte Carlo Dropout, I plot (see Figure 6.1) the reconstructed luminosity distance  $d_L(z)$  alongside its uncertainty bands and compare to the fiducial  $\Lambda$ CDM model.

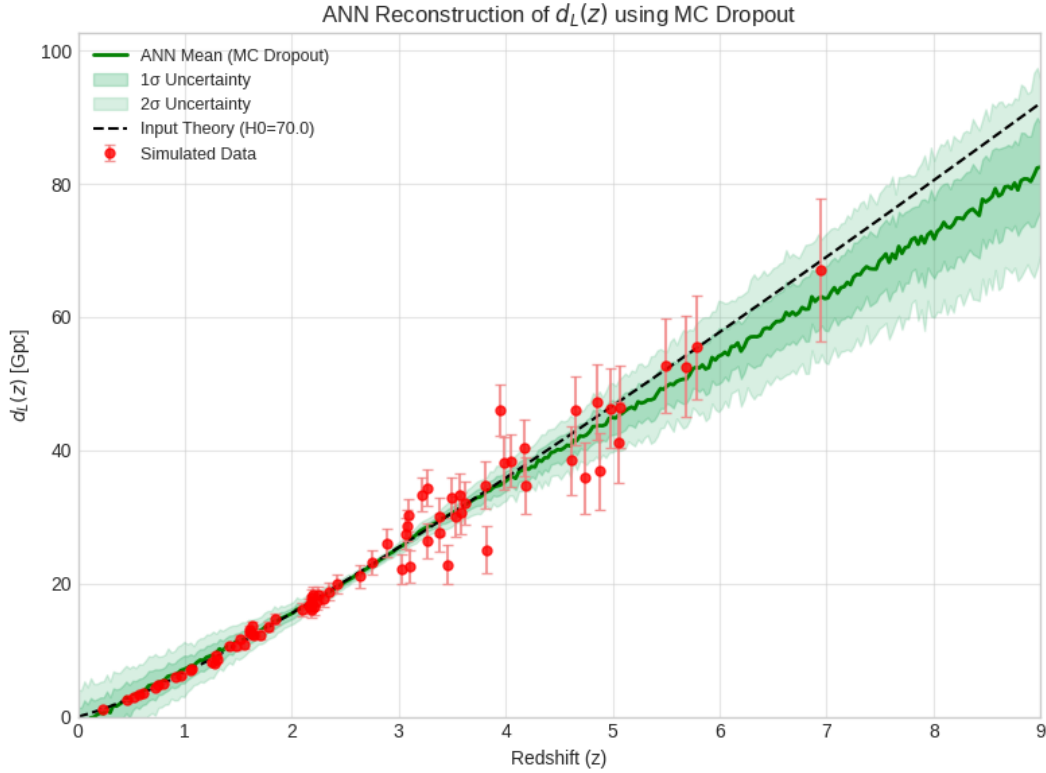


FIGURE 6.1: "No Delay" ANN mean prediction (green solid line) and MC Dropout uncertainty bands (shaded medium-seagreen) for  $d_L(z)$ . The dashed black line is the theoretical vanilla  $\Lambda$ CDM curve, and red points with error bars show the simulated training data.

### Plot Elements

- **ANN Mean:** The network's average output over multiple stochastic forward passes, plotted as a solid green curve.
- **Uncertainty Bands:**
  - 1  $\sigma$  band:  $\mu \pm \sigma$  shaded.
  - 2  $\sigma$  band:  $\mu \pm 2\sigma$  lightly shaded.

These are derived from the standard deviation of the MC Dropout ensemble.

- **Theory Curve:** Fiducial  $\Lambda$ CDM prediction  $d_L^{\text{theory}}(z)$  shown with a dashed black line.

- **Simulated Data:** Red markers with light-coral error bars indicating observational uncertainties in the training set.

#### Axes and Styling

- Redshift axis  $z$  spans the full training range up to  $z_{\max}$ .
- Distance axis  $d_L(z)$  in Gpc, starting at zero.
- Legend placed in the upper left, grid lines for readability.

**Interpretation** The ANN’s mean reconstruction closely follows the theoretical curve **even at high redshifts** unlike in the case for GPR, with the MC Dropout bands capturing the model uncertainty induced by limited training data and network regularization. Compared to Gaussian Process results (see Figure 5.4), the ANN shows slightly larger fluctuations at high redshift, highlighting areas for future improvement in derivative stability and uncertainty calibration.

**Future Directions** This initial ANN implementation successfully captures the nonlinear  $d_L(z)$  relationship. However, estimating  $H(z)$  requires precise derivatives of  $d_L(z)$ , which are highly sensitive to local oscillations in the ANN output. In future work, I plan to:

- Explore smooth architectures (e.g., DeepONets, spectral networks),
- Implement physics-informed neural networks (PINNs) with built-in derivative constraints,
- Compare MC Dropout uncertainty bands with ensemble and Bayesian Neural Networks.



## Chapter 7

# Conclusion

### 7.1 Conclusion

In this dissertation, I have demonstrated that Gaussian Process Regression (GPR) serves as a powerful, model-independent non-parametric framework for reconstructing the cosmic expansion history  $H(z)$  from gravitational-wave standard siren data. Applying GPR to simulated LISA catalogs, I recover  $H(z)$  with percent-level accuracy up to  $z = 4$ , and achieve uncertainties below 5% for  $z < 2$  when cross-checking against noise-free mean datasets. Although the ET analysis remains in progress due to the computational demands of MCMC on 1000-event catalogs, preliminary runs confirm that ET will yield constraints on  $H_0$  competitive with those from LISA.

I have also explored an Artificial Neural Network (ANN) as a rapid surrogate for luminosity-distance reconstruction. Although ANN reconstructions of  $d_L(z)$  achieve very low mean squared error ( $< 10^{-4}$ ), the lack of a straightforward uncertainty estimate and sensitivity to network architecture currently favor GPR for derivative-based inference such as computing  $H(z)$ .

#### 7.1.1 In a Nutshell

##### Progress

My key achievements to date include:

- **LISA Mock Catalogue GPR:** 100 astrophysically realistic mock catalogs generated for (80+1) events for each scenario: "No Delay", "Delay", and "Pop III". Stable reconstruction of  $H(z)$  up to  $z = 4$  with uncertainties  $\lesssim 5\%$  at  $z < 2$ ; cross-checked against noise-free mean catalogs.
- **ET Mock Data:** Full generation of 1000-event catalogs. initial GPR and MCMC runs completed, with final convergence pending additional compute time.
- **ANN for LISA:** Achieved  $d_L(z)$  reconstruction up to  $z = 9$  with mean squared error  $< 10^{-4}$  on validation sets, demonstrating feasibility of alternative machine-learning approaches.

##### Challenges

Several practical hurdles remain:

- **GaPP Incompatibility:** Legacy GaPP code to compute derivative via properties of GP kernels requires Python 2.7, necessitating a custom GPR implementation in Python 3.
- **Derivative Noise:** Finite-difference derivatives via `np.gradient` introduce high- $z$  scatter in  $H(z)$ .
- **Computational Cost:** Running GPR and MCMC on 1000-event ET catalogs is time-consuming; convergence demands significant CPU time.
- **GPU Limitations:** Restricted Google Colab GPU resources slow down deep-learning experiments, limiting ANN architecture exploration.

## 7.2 Future Work

Building on these results, several promising avenues can enhance and extend the methodology presented:

- **Additional Detectors:** Incorporate forecasts for next-generation observatories such as DECIGO (Mandel, Sesana, and Vecchio, 2018), the Big Bang Observer (BBO), Cosmic Explorer (CE) (Reitze et al., 2019), and TianQin, to evaluate their combined leverage on  $H(z)$ .
- **Dark Siren Analyses:** Develop statistical redshift inference techniques (e.g. galaxy catalog cross-correlations) to include dark sirens without electromagnetic counterparts, thereby multiplying the available event sample.
- **Advanced Machine Learning:** Investigate tools like LADDER (Shah et al., 2024) architectures for deep-learning reconstruction, which may better capture non-Gaussian features in noisy standard siren datasets.
- **Deep Gaussian Processes:** Combine Deep GPs with ANN outputs to enforce smoothness, improve extrapolation, and mitigate finite-difference noise in derivative estimation.
- **Ef-KAN Kernels:** Explore the recently proposed Ef-KAN kernel families for nonparametric inference (Cui et al., 2025), which offer adaptive, data-driven covariance structures that may further tighten constraints on  $H(z)$ .

By pursuing these directions, I aim to refine the non-parametric reconstruction toolkit for gravitational-wave cosmology and move closer to resolving the lingering Hubble tension in a fully model-independent manner.

## Appendix A

# Python Code

A

Below is the GitHub link for my repository, click the link to find the python implementation:

<https://github.com/Krishnanjan-Sil/IDEAS-TIH-ISIK-GP-2025>

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